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$$E \rightarrow E + T \mid T$$
$$T \rightarrow T \times F \mid F$$
$$F \rightarrow (E) \mid \mathbf{c}$$

- Convert the CFG G into an equivalent PDA, using the procedure described in Section 2.2. It suffices to give the compact form of the PDA, where we allow multiple symbols to be pushed onto the stack in a single step. (6 points)
- Describe the computational path by which the above PDA accepts the input string $c \times (c + c)$ (4 points)

Diagram illustrating a Turing Machine (TM) configuration with states and transitions:

- States: q_{start} , q_1 , q_2 , q_3 , q_4 , q_5 , q_6 , q_7 , q_8 , q_9 , q_{loop} , q_{accept} .
- Transitions (Input Σ , Output E):
 - $q_{start} \xrightarrow{\Sigma, \Sigma \rightarrow \$}$ q_1
 - $q_1 \xrightarrow{\Sigma, \Sigma \rightarrow E}$ q_{loop}
 - $q_2 \xrightarrow{\Sigma, \Sigma \rightarrow E}$ q_{loop}
 - $q_3 \xrightarrow{\Sigma, \Sigma \rightarrow T}$ q_{loop}
 - $q_4 \xrightarrow{\Sigma, \Sigma \rightarrow +}$ q_{loop}
 - $q_5 \xrightarrow{\Sigma, \Sigma \rightarrow E}$ q_{loop}
 - $q_6 \xrightarrow{\Sigma, T \rightarrow F}$ q_{loop}
 - $q_7 \xrightarrow{\Sigma, \Sigma \rightarrow T}$ q_{loop}
 - $q_8 \xrightarrow{\Sigma, \$ \rightarrow \Sigma}$ q_{loop}
 - $q_9 \xrightarrow{\Sigma, \Sigma \rightarrow E}$ q_{loop}
 - $q_{loop} \xrightarrow{\Sigma, \Sigma \rightarrow E}$ q_{loop} (Self-loop)
 - $q_{loop} \xrightarrow{\Sigma, \Sigma \rightarrow E}$ q_{accept}
- Additional transitions from q_{loop} to other states:
 - $q_{loop} \xrightarrow{\Sigma, F \rightarrow)}$ q_1
 - $q_{loop} \xrightarrow{\Sigma, \Sigma \rightarrow L}$ q_2
 - $q_{loop} \xrightarrow{\Sigma, E \rightarrow T}$ q_3
 - $q_{loop} \xrightarrow{\Sigma, T \rightarrow F}$ q_4
 - $q_{loop} \xrightarrow{\Sigma, F \rightarrow E}$ q_5
 - $q_{loop} \xrightarrow{\Sigma, E \rightarrow L}$ q_6
 - $q_{loop} \xrightarrow{\Sigma, L \rightarrow E}$ q_7
 - $q_{loop} \xrightarrow{\Sigma,) \rightarrow \Sigma}$ q_8
 - $q_{loop} \xrightarrow{\Sigma, X \rightarrow \Sigma}$ q_9
 - $q_{loop} \xrightarrow{\Sigma, F \rightarrow E}$ q_{loop}
- Handwritten notes:
 - Bottom left: "bottom line not left" with an arrow pointing to the bottom row of transitions.
 - Right side: A list of steps:
 1. $\rightarrow start$
 2. $q_{start} \rightarrow q_1$
 3. $q_1 \rightarrow q_{loop}$
 4. $q_{loop} \rightarrow q_1$
 5. $q_{loop} \rightarrow q_2$
 6. $q_{loop} \rightarrow q_3$

parenting
NOT letter
K

Step	Stack	String
1. $\rightarrow q_{start}$	[]	ϵ
2. $q_{start} \rightarrow q_1$	[i]	i
3. $q_1 \rightarrow q_{loop}$	[i e]	ie
4. $q_{loop} \rightarrow q_{loop}$	[i T]	iet
5. $q_{loop} \rightarrow q_c \rightarrow q_s$ $\rightarrow q_{loop}$	[T F F] [c c f]	ietffccf
6. $q_{loop} \rightarrow q_{loop}$	[n n n]	ietffccfn
7. $q_{loop} \rightarrow q_{loop}$	[]	ietffccfn
8. $q_{loop} \rightarrow q_{loop}$	[f]	ietffccfnf
9. $q_{loop} \rightarrow q_c \rightarrow q_s$ $\rightarrow q_{loop}$	[c c f] [e e f]	ietffccfnfecfe
10. $q_{loop} \rightarrow q_{loop}$	[e f f] [e n n]	ietffccfnfecfen
11. $q_{loop} \rightarrow q_{loop}$	[]	ietffccfnfecfen
12. $q_{loop} \rightarrow q_{loop}$	[T T T]	ietffccfnfecfnTTT
13. $q_{loop} \rightarrow q_{loop}$	[f T T] [c c f]	ietffccfnfecfnTTTcf
14. $q_{loop} \rightarrow q_{loop}$	[f T T] [e n n]	ietffccfnfecfnTTTcfenn
15. $q_{loop} \rightarrow q_{loop}$	[]	ietffccfnfecfnTTTcfenn
16. $q_{loop} \rightarrow q_{loop}$	[c c f]	ietffccfnfecfnTTTcfenncfc
17. $q_{loop} \rightarrow q_{loop}$	[e n n]	ietffccfnfecfnTTTcfennecne
18. $q_{loop} \rightarrow q_{loop}$	[]	ietffccfnfecfnTTTcfennecne
19. $q_{loop} \rightarrow q_{loop}$	[f]	ietffccfnfecfnTTTcfennecnff
20. $q_{loop} \rightarrow q_{loop}$	[]	ietffccfnfecfnTTTcfennecnff
21. $q_{loop} \rightarrow q_{accept}$	[]	ietffccfnfecfnTTTcfennecnff

[illegible]

Sorry, this got messy really fast...

2. Let B be the language of all palindromes over $\{c, d\}$ containing an equal number of c 's and d 's. Show that B is not context-free using the pumping lemma. (10 points)

Note that a string $w \in \{c, d\}^*$ belongs to B if and only if (a) w is a palindrome, and (b) w has an equal number of c 's and d 's.

$B = \{c, d\}$ palindromes where $|c| = |d|$

strings $\in B$: ε , $cddc$, $dc cd$, $c^i d^{2i} c^i$, $d^i c^{2i} d^i$, $c^i d^i$

Proof:

Assume B is a context free language. There must be a pumping length p such that $|s| \geq p$. Choosing string $s = d^p c^2 p d^p$, by the pumping lemma for non context free languages, s can be split into $uvxyz$ satisfying:

1. for each $i \geq 0$, $uv^i xy^i z \in A$
2. $|vy| > 0$
3. $|vxy| \leq p$

There are 5 cases for the pumping of a substring vxy

$\underbrace{d d d d}_{1} \underbrace{c c c c c c c c}_{2} \underbrace{d d d d}_{3}$

1. entirely in left d^p block $\Rightarrow$ pumping results in non-palindrome contribution
2. entirely in middle $c^2 p$ block $\Rightarrow$ pumping results in $|c| \neq |d|$ contribution
3. entirely in right d^p block $\Rightarrow$ pumping results in non-palindrome contribution
4. contains $d^2 c^2$ $\Rightarrow$ pumping results in non-palindrome contribution
5. contains $c^m d^n$ $\Rightarrow$ pumping results in non-palindrome contribution

This having shown that there is no way to pump string $s = d^p c^2 p d^p$ such that it is included in B , by the pumping lemma, B must be a non-context free language.

3. Let $\Sigma = \{1, 2, 3, 4\}$, and

$$C = \left\{ w \in \Sigma^* \mid \begin{array}{l} \text{in string } w, \text{ the number of 1s equals the number of 2s,} \\ \text{and the number of 3s equals the number of 4s} \end{array} \right\}$$

Show that C is not context-free using the pumping lemma. (10 points)

$$C = \{ w \in \Sigma \mid |1| = |2| \text{ \& } |3| = |4| \}$$

strings: $\epsilon, 1234, 12, 34, 1122344, 1^6 2^6 3^4 4^4$

Proof:

Assume that C is a context free language. Then, there exists a pumping length p such that $|s| \geq p$. Choosing the string $s = 1^p 2^p 3^p 4^p$, by the pumping lemma, s can be split into $uvxy$ satisfying:

1. for each $i \geq 0$, $uv^i xy^i z \in A$
2. $|vy| > 0$
3. $|vxy| \leq p$

1 1 1 2 2 2 2 3 3 3 3 4 4 4 4

There are 2 cases for subdividing $uvxy$

* without loss of generality *

1. entirely in $1^p, 2^p, 3^p$, or 4^p block $\Rightarrow$ pumping results in either $|1| \neq |2|$ or $|3| \neq |4|$
2. spans boundary $1^p 2^p, 2^p 3^p$, or $3^p 4^p \Rightarrow$ pumping also results in a failed equality
or either $|1| \neq |2|$ or $|3| \neq |4|$

contradiction

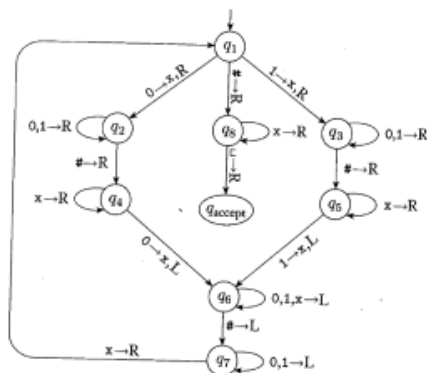
contradiction

Having shown that there is no way to pump $s = 1^p 2^p 3^p 4^p$ such that it belongs to C , then C must be a non-context free language.

4. The exercise concerns the Turing Machine (TM) in Example 3.9 of the text, which is reproduced below. Note that the reject state is not explicitly shown in the state diagram, but is the default option for any transition not shown in the state diagram; assume that the head moves right when transitioning to the reject state.

The following is a formal description of $M_1 = (Q, \Sigma, \Gamma, \delta, q_1, q_{\text{accept}}, q_{\text{reject}})$, the Turing machine that we informally described (page 167) for deciding the language $B = \{w\#w \mid w \in \{0,1\}^*\}$.

- $Q = \{q_1, \dots, q_8, q_{\text{accept}}, q_{\text{reject}}\}$,
- $\Sigma = \{0, 1, \#\}$, and $\Gamma = \{0, 1, \#, x, \epsilon\}$.
- We describe δ with a state diagram (see the following figure).
- The start, accept, and reject states are q_1 , q_{accept} , and q_{reject} , respectively.



In each of the configurations below, give the sequence of configurations that the TM enters when started on the indicated input string. (10 points)

- (a) 0#0
(b) 0##0
(c) 01#00
(d) 01#01

see next page

a) 0#0

Transition	Action	Configuration
$q_1 \rightarrow q_2$	$0 \rightarrow x, R$	$\overset{?}{\cancel{x}} \# 0 \sqcup . . .$
$q_2 \rightarrow q_4$	$\# \rightarrow R$	$x \overset{?}{\#} 0 \sqcup . . .$
$q_4 \rightarrow q_6$	$0 \rightarrow x, L$	$x \# \overset{?}{\cancel{0}} \sqcup . . .$
$q_6 \rightarrow q_7$	$\# \rightarrow L$	$x \overset{?}{\#} x \sqcup . . .$
$q_7 \rightarrow q_1$	$x \rightarrow R$	$\overset{?}{x} \# x \sqcup . . .$
$q_1 \rightarrow q_9$	$\# \rightarrow R$	$x \overset{?}{\#} x \sqcup . . .$
$q_9 \rightarrow q_8$	$x \rightarrow R$	$x \overset{?}{\#} x \sqcup . . .$
Accept!	$\sqcup \rightarrow R$	$x \# x \overset{?}{\sqcup} . . .$

b) 0

Transition	Action	Configuration
$q_1 \rightarrow q_2$	$0 \rightarrow x, R$	$\downarrow$ 0 # # 0
$q_2 \rightarrow q_4$	$\# \rightarrow R$	$\downarrow$ x # # 0
$q_4 \rightarrow q_{\text{reject}}$	REJECT	$\downarrow$ x # # 0

There is no transition
from state 1 on '0';
reject string
0##0

c) 01#00

Transition	Action	Configuration
$q_1 \rightarrow q_2$	$0 \rightarrow x, R$	$\overset{1}{\cancel{x}} \ 1 \ # \ 0 \ 0 \ \sqcup \ \dots$
$q_2 \rightarrow q_3$	$0, 1 \rightarrow R$	$x \ 1 \ # \ 0 \ 0 \ \sqcup \ \dots$
$q_3 \rightarrow q_4$	$\# \rightarrow R$	$x \ 1 \ # \ 0 \ 0 \ \sqcup \ \dots$
$q_4 \rightarrow q_5$	$0 \rightarrow x, L$	$x \ 1 \ # \ \overset{1}{\cancel{x}} \ 0 \ \sqcup \ \dots$
$q_5 \rightarrow q_6$	$\# \rightarrow L$	$x \ 1 \ # \ x \ 0 \ \sqcup \ \dots$
$q_6 \rightarrow q_7$	$0, 1 \rightarrow L$	$x \ 1 \ # \ x \ 0 \ \sqcup \ \dots$
$q_7 \rightarrow q_8$	$x \rightarrow R$	$x \ 1 \ # \ x \ 0 \ \sqcup \ \dots$
$q_8 \rightarrow q_9$	$1 \rightarrow x, R$	$x \ \overset{1}{\cancel{x}} \ # \ x \ 0 \ \sqcup \ \dots$
$q_9 \rightarrow q_{10}$	$\# \rightarrow R$	$x \ x \ # \ x \ 0 \ \sqcup \ \dots$
$q_{10} \rightarrow q_{11}$	$x \rightarrow R$	$x \ x \ # \ x \ 0 \ \sqcup \ \dots$
$q_{11} \rightarrow \text{reject}$	REJECT	$x \ x \ # \ x \ 0 \ \sqcup \ \dots$

no transitions from
 q_{11} reading 0;
reject string

01#00

d) 01#01

Transition	Action	Configuration
$q_1 \rightarrow q_2$	$0 \rightarrow x, R$	$\downarrow$ 0 1 # 0 1 u ...
$q_2 \rightarrow q_3$	$0, 1 \rightarrow R$	$\downarrow$ x 1 # 0 1 u ...
$q_3 \rightarrow q_4$	$\# \rightarrow R$	$\downarrow$ x 1 # 0 1 u ...
$q_4 \rightarrow q_5$	$0 \rightarrow x, L$	$\downarrow$ x 1 # 0 1 u ...
$q_5 \rightarrow q_6$	$\# \rightarrow L$	$\downarrow$ x 1 # x 1 u ...
$q_6 \rightarrow q_7$	$0, 1 \rightarrow L$	$\downarrow$ x 1 # x 1 u ...
$q_7 \rightarrow q_8$	$x \rightarrow R$	$\downarrow$ x 1 # x 1 u ...
$q_8 \rightarrow q_9$	$1 \rightarrow x, R$	$\downarrow$ x 1 # x 1 u ...
$q_9 \rightarrow q_{10}$	$\# \rightarrow R$	$\downarrow$ x x # x 1 u ...
$q_{10} \rightarrow q_{11}$	$x \rightarrow R$	$\downarrow$ x x # x 1 u ...
$q_{11} \rightarrow q_{12}$	$0 \rightarrow x, L$	$\downarrow$ x x # x 0 u ...
$q_{12} \rightarrow q_{13}$	$0, 1, x \rightarrow L$	$\downarrow$ x x # x x u ...
$q_{13} \rightarrow q_{14}$	$\# \rightarrow L$	$\downarrow$ x x # x x u ...
$q_{14} \rightarrow q_{15}$	$x \rightarrow R$	$\downarrow$ x x # x x u ...
$q_{15} \rightarrow q_{16}$	$\# \rightarrow R$	$\downarrow$ x x # x x u ...
$q_{16} \rightarrow q_{17}$	$x \rightarrow R$	$\downarrow$ x x # x x u ...
$q_{17} \rightarrow q_{18}$	$\epsilon \rightarrow R$	$\downarrow$ x x # x x u ...
$q_{18} \rightarrow q_{accept}$	$\sqcup \rightarrow R$	$\downarrow$ x x # x x u ...

Accept!